

1 Challenge

Naive RAG

Vector / No-LLM

Mem0

Vector / LLM

LangMem

Vector / LLM

Graphiti

Graph / LLM

Letta

Hierarch. / LLM

Architecturally heterogeneous
→ incomparable without a standard

2 Gap

Existing benchmarks:

measure task success

memory = black box

Missing:

Write latency?

Recall / omission?

Update consistency?

User isolation?

No real-conversation dataset
for system-level evaluation

3 Our Approach

MemDialogue Dataset

10,000 real-conversation memory events

from LMSYS-Chat-1M

triplet: raw_text / query / ground_truth

MemSysBench Evaluation

RQ1 Write Efficiency

RQ2 Retrieval Quality

RQ3 Scalability

RQ4 Temporal Consistency

RQ5 Isolation & Privacy

RQ6 Portability

→ **13 empirical findings**